

# Friction: test the surfaces

Autumn 1 · Week 3 · Lesson B · 40 minutes

## Learning objective

Explain friction's effects using a controlled three-surface comparison and repeated stopping-distance evidence.

## Evidence to look for

Record actual participation/support and whether the learner identified controls, recorded or interpreted distances and justified a qualified conclusion. Observation/directing is not excluded; model-data interpretation is identified honestly.

## Source lesson timing

| Elapsed minutes | Source stage        | Minutes |
|-----------------|---------------------|---------|
| 0–32            | Independent Work    | 32      |
| 32–36           | Review the evidence | 4       |
| 36–40           | Exit Ticket         | 4       |

Split slide times are added within their original source stages. Printed tasks and adaptations replace activity within these timings; they do not add extra lesson time.

## Equipment and preparation

- Adult-secured low ramp and one small sliding block
- Three level surface samples with a stable ramp join
- Ruler or tape, fixed zero/edge markers, tray or padded stop
- Lesson A prediction and pupil record; optional offline XLSX for actual measurements

**TEACHING SEQUENCE**

# Guide the lesson

Use the source stage times; choose prompts appropriate to the pupil.

**Independent Work · 32 minutes**

Retrieve and set up 4 min; test three surfaces with two trials each 20 min; pause and check 2 min; repeat one concern, conclude and tidy 6 min. Alternative uses explicitly labelled source model data within the same time.

**Review the evidence · 4 minutes**

Revisit one conclusion against the existing results or labelled model data. Engage with the response and adapt the explanation or scaffold. Preserve original values; no separate loop sheet is needed.

**Exit Ticket · 4 minutes**

Choose a supported, standard or stretch friction explanation; attempt first, then check and improve.

**TEACHER ANSWERS**

# Worked answers and expected evidence

Use after an attempt. Accept equivalent correct explanations; record support given.

## Model-data conclusion

Source invented data: card 42/44 cm, fabric 25/27 cm, rubber mat 15/18 cm. Rubber mat produced the shortest run-outs in both trials. This supports greater resistance to sliding in the stated setup, not a universal ranking. No means required.

### 1. Prediction and controls

A prediction is justified rather than marked right/wrong. Accept names of available surfaces with reasoning about resistance at the contact. Controls: same block, low ramp height/angle, release mark/no push, level samples, ramp join, fixed measuring zero and same block edge. Keep original prediction for comparison.

### 2. Three surfaces, repeated evidence

No fixed real-data answer: check surface identities, two honestly recorded trials where performed, cm units and relevant controls. A non-slide is not silently converted to 0 cm; identify what happened. Shorter run-out distance supports greater resistance to sliding for this block/setup if relevant controls are maintained. Accept uncertainty or no clear separation.

### 3. Alternative: model-data interpretation

Rubber mat 15/18 cm is shorter than fabric 25/27 cm and card 42/44 cm. This supports greater resistance to sliding for the stated block/setup if block, release, ramp, join and zero stayed fixed. Material behaviour depends on the actual contacting surfaces and conditions. This is supplied model-data analysis, not the learner's measurements.

### 4. Check, improve and conclude

Look for one concrete concern, e.g. pushed block, shifted surface or zero. Difference alone is not proof of error. Retrieved value remains separate and original retained. Real conclusion must fit actual evidence; model conclusion can use 15/18 cm versus 25/27 cm and 42/44 cm with setup qualification. Suitable improvement: fixed no-push release or clearer zero, not 'choose numbers that match'.

### 5. Review the evidence and exit

Review: check the conclusion against both trials and the controls; retain uncertainty honestly. Exit: brakes/shoes/tyres/handles; brake contact resists sliding and slows the wheel, with tyre-road grip also needed; contrast this with unwanted engine heating/wear. Accept equivalent responses.

## Misconceptions and corrections

---

- Friction is always bad: its effects depend on the job.
- Smooth means no friction: smooth surfaces can still grip.
- Shorter distance directly measures a friction force: distance provides comparative evidence under controls, not a force reading in N.
- A surprising result must be deleted: keep it and note a specific method concern if known.

# Teach safely; keep the evidence

The source record identifies this conversion; it is not a claim of publication.

## Safety

---

- Use a low adult-secured ramp and light block; clear run and padded stop.
- Hands stay outside the run; agree release signal and no pushing.
- No oils, loose powders, wet floors or pupils sliding on floors.
- Offer quiet observer/director/recorder roles or labelled no-kit data interpretation.

## Access and offline alternatives

---

- Speak, sign, point, draw, dictate or direct instead of manipulating equipment.
- If kit unavailable use the source invented teaching data and label the activity interpretation.
- No HTML, video, online account, photograph or public sharing needed.

## Close and continue

---

Finish at 40 minutes: 32 workshop + 4 evidence review + 4 exit. Next lesson: Levers, pulleys and gears.

## Source and revision

---

Repository: [MattRoper1977/Lessons](https://github.com/MattRoper1977/Lessons)

Source path: [Science\\_Teesside/Grow/SCI\\_G\\_W3\\_Friction.html](#)

Historical conversion source Git blob: [170d772a5e9d23909e95a534c5c1e16ee7df67c8](#)

Current source link (mutable): [https://github.com/MattRoper1977/Lessons/blob/main/Science\\_Teesside/Grow/SCI\\_G\\_W3\\_Friction.html](https://github.com/MattRoper1977/Lessons/blob/main/Science_Teesside/Grow/SCI_G_W3_Friction.html)

Feedback revision: 2026-09-15 · draft pilot

## Recorded conversion changes

- Workshop remains 32 minutes, followed by four-minute evidence review and four-minute exit. The final page gives integrated feedback guidance; no separate participation stage or loop record is required.
- Paper records and adult demonstration replace HTML controls; observation, direction and dictation are equivalent routes.
- The source's invented teaching data remain clearly identified as model data, never pupil measurements.
- Surface conclusions apply to the tested block and setup, not universal material rankings.
- No publisher footage copied; exact original logo remains for release coordinator to resolve.

# Feedback and marking · Friction

## Make participation possible

Use regulation time, a trusted adult, accessible response routes and permission to refuse. The small help prompt is optional; useful feedback remains part of ordinary teaching. Refusal is information, not a reason to invent a response or punish passing.

## Lesson A: a claim worth revisiting

During We Do 2, invite a pupil to identify one word or idea in a sorted claim that needs checking. Hear, read or watch the response. Use the actual surface-pair example to adapt the explanation, scaffold or next question. Keep this within the existing eight minutes of sorting and correction, then use two minutes for the exit.

## Lesson B: review the evidence

After the 32-minute workshop, use four minutes to revisit one conclusion and its supporting results with the adult, then four minutes for the exit. Use the existing record or an equivalent response; do not require a separate loop sheet. Explain any limit on a requested change and check whether the response helped.

## Listen and act

Space makes participation possible; Voice is the pupil's response in their own modality; Audience requires the adult to engage with it; Influence is the resulting adjustment to explanation, task, scaffold or subsequent teaching. One optional displayed prompt does not limit genuine feedback exchanges.

## Use marking codes honestly

Use R only after a pupil response and genuine adult engagement. A click, download, opened explanation or completed exit does not automatically establish Audience, Influence, R or an assessment outcome. Record only the evidence and help actually provided; do not require a duplicate written record for every task. Use agreed policy codes appropriately. Extended writing or assessment may need one or two precise next steps and response time.

## Policy and timing

Feedback & Marking Policy 2025/2026 (Pilot), Issue 1, May 2026; master order §25.3. Lesson A remains 40 minutes. Lesson B remains 32 + 4 + 4 minutes. No extra pupil form, signature, standalone participation stage or progression gate is required.